

Principal-timestep Restricted Init via Sparse Matrix-decomposition in Flow-matching

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Abstract

Flow-matching diffusion models have recently emerged as a strong paradigm for high-fidelity visual generation. However, their prohibitively high fine-tuning cost limits scalability to downstream tasks. While Low-Rank Adaptation (LoRA) combined with spectral initialization has demonstrated accelerated convergence and improved performance in autoregressive language models by better aligning gradient directions, we find that it fails to deliver similar gains in diffusion fine-tuning, often yielding marginal or even negative improvements over vanilla LoRA. We attribute this discrepancy to a fundamental mismatch between LoRA’s low-rank parameterization and the intrinsically high-rank gradients induced by the flow-matching objective. In particular, stochastic timestep sampling introduces directionally heterogeneous gradient signals across training steps, leading to misaligned updates under low-rank constraints. To address this issue, we propose PRISM-LoRA, a Principal-timestep Restricted Init via Sparse Matrix-decomposition framework that improves gradient alignment during fine-tuning. Our method consists of two key components: (i) principal timestep selection, which restricts initialization gradients to a subset of dominant timesteps to suppress effective gradient rank, and (ii) principal channel filtering, which removes task-irrelevant channels, enabling the one-step spectral initialization gradient to better align with the long-horizon optimization trajectory. Extensive experiments demonstrate that PRISM-LoRA consistently improves both convergence speed and final performance across multiple diffusion fine-tuning benchmarks, including subject-driven generation, controllable generation, and deblurring, achieving not only performance improvement but also earlier stages of convergence over baseline LoRA and other spectral-init methods. Our code is available at <https://anonymous.4open.science/r/Prism-LoRA-28ED>.

1 Introduction

Recently, the spectral initialization on LoRA has been receiving increasing research attention. Other LoRA variants [30, 23, 33], either require a dedicated design of hyperparameters or extra training parameters to improve the performance of fine-tuning. In contrast, spectral initialization [39, 47, 28] requires few hyperparameters and comes with tight theoretical guarantees on fine-tuning scaling. By initializing the LoRA adapters with the decomposition of either the pre-trained weights or a gradient obtained from a single-batch forward pass, it achieves both stronger performance and faster convergence.

The achievement of spectral initialization can be attributed to the successful transfer of the knowledge from the fine-tuning pattern of the full-parameter model to the low-rank parameters of LoRA.

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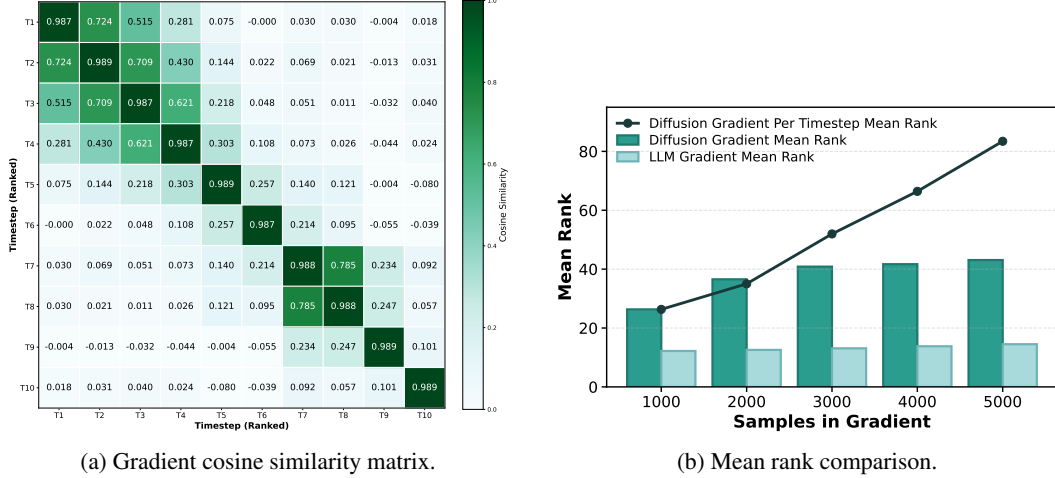


Figure 1: Left figure shows the cosine similarity matrix on different diffusion timesteps, right figures show when sampled from a wider range of timesteps and sampling numbers, the difference in gradient rank and compared with LLM gradient.

Existing one-step gradient methods typically justify their effectiveness along two axes: (i) they show that the LoRA parameterization obtained by decomposing the first-step gradient attains the minimal approximation error relative to full fine-tuning at initialization; and (ii) they argue that this initialization remains well-aligned with the dominant gradient directions throughout subsequent training. Both arguments implicitly rely on a critical premise—that the *effective rank* of the full-model gradient is comparable to the LoRA rank. As effective rank is related to the energy distribution towards singular value, it indicates that using how many of the singular vectors on the front would have a reconstructive effect towards the original matrix. If effective rank matches LoRA’s rank, a good LoRA initialization weight derived from the decomposition result of the matrix can set an optimization start point similar as a full fine-tuning model does.

This premise, however, breaks down in the diffusion setting. Figure 1 illustrates the issue from two angles. Figure 1a shows that gradients from different timesteps share little similarity, indicating that they span largely distinct subspaces. Consequently, the aggregated gradient $G^{\mathfrak{A}} = \mathbb{E}_t[G_t]$ exhibits a substantially higher effective rank than any single G_t . Figure 1b quantifies this effect. We define the empirical effective rank as the number of top singular vectors needed to reconstruct 99% of the gradient’s Frobenius norm, and progressively enlarge the aggregated timestep range (“1000” covers T_1-T_2 , “2000” covers T_1-T_4 , and so on). Compared to an LLM baseline, the diffusion gradient’s effective rank grows markedly faster as more timesteps are aggregated; the overlaid line further shows that this rank itself rises as timesteps move from noise to data. As a result, $G^{\mathfrak{A}}$ inherits an inflated rank structure, and the low-rank SVD on which one-step methods like LoRA-One rely becomes a lossy operation in the diffusion setting.

We attribute this failure to a fundamental **mismatch between spectral-init’s low-rank assumption and the gradient geometry of diffusion training**. In LLMs, the one-step gradient is well-approximated as low-rank, justifying spectral initialization. In diffusion, the one-step gradient aggregates over a sequence of timesteps $\mathbb{E}_t[G_t]$; since different t correspond to denoising tasks at different signal-to-noise ratios, the G_t point in widely divergent directions. The aggregated gradient, therefore, has substantially higher effective rank, violating the core premise of spectral initialization.

On this basis, we raise 2 hypotheses corresponding to 2 aspects, shown as follows:

Hypothesis 1 (Principal Timesteps reduce effective rank) Restricting gradient estimation to small timestep t (closer to noise) $t \rightarrow 0$ yields a gradient matrix $G^{\mathfrak{A}}$ with substantially lower effective rank than uniform timestep sampling, improving the fidelity of ranking- r spectral initialization.

Hypothesis 2 (Sparse decomposition improves finite-sample subspace recovery) The irrelevant channel signal would hinder the reconstructed weight from being aligned with the fine-tuning orientation.

Corresponding to Hypothesis 1, we introduce a principal timestep selection strategy that confines gradient estimation to a small set of timesteps, suppressing the effective rank of the guidance gradient before decomposition. Corresponding to Hypothesis 2, we apply a sparse channel decomposition that isolates task-relevant channels from the gradient signal, so that the reconstructed low-rank update aligns with the long-horizon fine-tuning trajectory rather than being polluted by irrelevant directions. Our contributions are summarized as follows:

1. We identify and analyze the failure mode of spectral initialization in diffusion fine-tuning, attributing it to a mismatch between the low-rank assumption of spectral initialization and the high-rank gradient geometry induced by stochastic timestep sampling.
2. Motivated by two hypotheses on gradient rank and channel relevance, we propose PRISM-LoRA, which combines principal timestep selection and sparse channel decomposition to produce a low-rank, well-aligned initialization for diffusion LoRA.
3. We validate our method on four downstream tasks (Dreambooth, Canny, Depth, Deblur) across Stable Diffusion-3 and Flux-1, achieving not only performance improvement but also training acceleration at the initial training stage over strong spectral-init baselines.

2 Related Work

Diffusion-based Generative Models. The diffusion model was originally introduced by [34], which has replaced Generative Adversarial Network (GAN)-based methods [38, 43, 24], and has been widely applied in the field of image synthesis [14, 8, 46]. Compared to adversarial optimization approaches, which easily lead to training instability and even collapse, diffusion models utilize the probabilistic optimization scheme to ensure stable convergence and generation diversity. The Latent Diffusion Model (LDM) [31] reduces computational demands by transferring the diffusion process from pixel space to latent space. Based on it, Transformer-based architectures such as Diffusion Transformer (DiT) [29, 51, 4, 17, 1], Stable Diffusion 3.0/3.5 [7], FLUX [18] further extend diffusion’s capacity to model long-range dependencies and scale to massive multi-modal datasets. Meanwhile, Rectified Flow optimization [19, 21, 22] enables faster sampling with fewer denoising steps without degrading quality. Although the merits are obvious, pre-training a diffusion model is undoubtedly expensive.

Post-training of Diffusion Models. Compared to pre-training, fine-tuning the fully-trained diffusion model on downstream task with a small-scale dataset [42, 16, 5] can be more feasible and cost-friendly. DreamBooth [32] finetunes text-to-image diffusion models using a few subject-specific images provided by practitioners, binding the subject to a unique identifier token to enable high-fidelity and controllable personalized generation. [44, 26, 27] freeze the denoising backbone and finetune an additional module to transfer conditional controls into the noise latent space, thereby achieving spatially-aligned generation. Meanwhile, due to the decoupled design with the denoising network—parameter updation occurring only on the trainable adapter, the training stability and convergence speed are both superior to the full-parameter fine-tuning. To further compress the trainable parameters, [35, 36, 48, 37, 9] introduce efficient LoRA module [13] for diffusion fine-tuning, which not only reduces the demand for gpu memory, but also makes it convenient for the diffusion model to switch freely between different design tasks. However, the number of update steps required for fine-tuning is still quite large, and the performance is difficult to match that of full-parameter fine-tuning.

Improving Fine-tuning Efficiency. Low-Rank Adaptation (LoRA) [13, 6, 23, 33] has dominated efficient large vision/language models’ fine-tuning. Apart from the parameter efficiency, researchers have also extended to explore time efficiency. On the one hand, some research works focus on improving initialization strategies to enhance the effectiveness of low-rank updates. [28, 39, 47] propose an initialization method based on gradient update approximation, which enables the low-rank parameters to approach the optimal update direction at the early stage of training. [52, 25, 20] leverage singular value decomposition (SVD) [40] to confine the adaptation within the principal singular subspace, thereby enabling more stable and robust initialization and updates of model weights by better aligning with the intrinsic structure of the pretrained model. On the other hand, there are also efforts to improve training efficiency by focusing on the optimization process. [10, 15, 45, 49] allocate different learning rates to different matrices in low-rank decomposition, significantly accelerating convergence and improving performance without changing the initialization method. However, due

to the multi-optimization training objectives in diffusion fine-tuning, these methods fail to capture a robust enough gradient by only relying on one-step gradient to guide the overall training process.

3 Method

In this section, we will introduce how we solve the new problem of the gradient-based method applied in the diffusion domain. We understand that fluctuating training loss is the inevitable nature of fine-tuning the diffusion model, and it can hardly ensure that the 1-step optimization direction points at the average optimization direction of the overall fine-tuning process. We notice this problem and simplify it in Sec. 3.1, defined as *High-ranking $G^\natural$ problem*. It can be solved in 2 aspects, the timestep and channel, shown in Sec. 3.2 and Sec. 3.3, respectively. Finally, we formulate the overall training process in Sec. 3.4, to have an overall look at our method. For preliminaries, see Appendix. A.

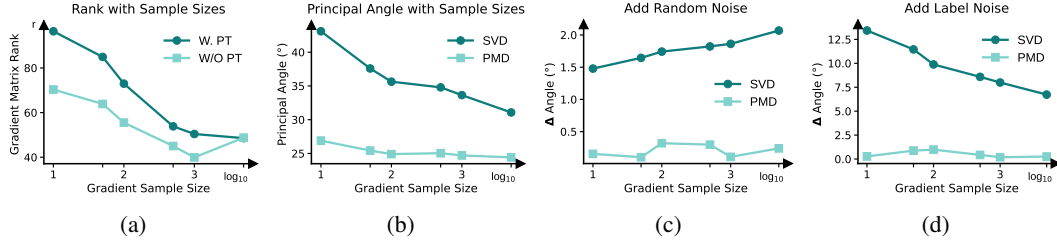


Figure 2: **Empirical validation of our proposed hypotheses.** Fig. 2a indicates that the principal timestep(PT) achieves a lower rank than stochastic timesteps sampling. Fig. 2b under principal timestep sampling, calculates the principal angle with G with $50K$ sample size, achieving a closer direction. Fig. 2c and 2d shows golden channel module has a better ability in filtering out irrelevant channel information compared with SVD. For further analysis, see Sec. 3.3.

3.1 Barriers in Flow-matching Diffusion Model

High-ranking $G^\natural$. Near-orthogonal per-timestep gradients do not lead to a low-rank average, as each G_t contributes an independent direction to the column or row space of $G^\natural$. Formally, if G_{t_1} and G_{t_2} have near-orthogonal top- r subspaces, the rank of their sum approaches $\text{rank}(G_{t_1}) + \text{rank}(G_{t_2})$. Averaging over a continuous number of timesteps therefore inflates the effective rank of $G^\natural$ relative to any single G_t . From a mathematical perspective, average gradient from different directions would lead to a high-ranking $G^\natural$. It has 2 drawbacks: The decomposed LoRA parameter from high-ranking $G^\natural$ has a conflict with the low-ranking training objective with LoRA; High-ranking $G^\natural$ may have a deviated guidance on multiple training objects. However, directly measuring the true rank of the gradient matrix is challenging. To address this, we convert this problem into measuring the spectral of the gradient matrix, which would have a positive correlation with the actual rank r^* of $G^\natural$.

From gradient rank to posterior covariance spectrum. Directly analyzing the spectrum of $G^\natural$ is intractable, as it entangles the data-driven residual with the architecture-dependent Jacobian J_θ . Under a locally linearized view, the per-timestep gradient covariance decomposes as

$$G_t \approx \mathbb{E}_{x_t} [J_\theta^\top \Sigma_v(t) J_\theta], \quad (1)$$

where $\Sigma_v(t) = \frac{1}{t^2} \text{Var}[x_0 | x_t]$ is the velocity-regression residual covariance. Since the architecture-dependent factor J_θ admits no general treatment, and since an ill-conditioned output target necessarily induces an ill-conditioned G_t , we adopt $\Sigma_v(t)$ as a principled proxy for the spectral behavior of $G^\natural$. Furthermore, as absolute rank is numerically unstable, we replace it with a scale-invariant surrogate—the concentration of the spectrum—which is positively correlated with the effective rank and amenable to rigorous comparison via majorization.

We measure spectral concentration via the r_N metric: the minimum number of top eigenvalues needed to capture $N\%$ of the total variance, $r_N(\Sigma) = \min\{r : \sum_{i=1}^r \sigma_i \geq N\%\}$ where σ_i are singular values. Small r_N indicates a sharp, concentrated spectrum (effectively low-rank); $r_N \rightarrow d$ indicates a flat spectrum (high-rank).

3.2 Principal Timestep

The effectiveness of the principal timestep is based on the following justification: the spectrum of the one-step gradient covariance $\Sigma_v(t)$ becomes flatter when mixing with more close-to-target timestep. As a result, restricting spectral initialization to smaller t leads to top- r subspace capturing a substantially larger fraction of the total signal. We formulate this theorem as follows:

Theorem 1 (Monotone spectral flattening). *Under the assumptions of Sec. 3.1, the rank- N index r_N of the one-step gradient covariance $G^\natural$ is monotonically non-decreasing in t on $[0, 1]$:*

$$r_N(\Sigma_v(0)) \leq r_N(\Sigma_v(0.5)) \leq r_N(\Sigma_v(1)) = \lceil N/100, d \rceil. \quad (2)$$

By the flow-matching identity $x_t = (1-t)x_0 + tx_1$, we eliminate x_1 via $x_1 = (x_t - (1-t)x_0)/t$, so that the posterior covariance $\text{Var}[x_0 | x_t]$ depends on x_0 alone. Diagonalizing in the eigenbasis of Σ_1 , the i -th singular value of $\Sigma_v(t)$ admits the closed form

$$\sigma_i(t) = \frac{\lambda_i}{(1-t)^2 + t^2\lambda_i}, \quad (3)$$

where $\{\lambda_i\}_{i=1}^d$ are the singular values of the data covariance $\Sigma_1 = \text{Cov}(x_1)$, and by symmetric positive semi-definiteness coincide with its eigenvalues. A full derivation is deferred to Appendix B.1.

Two-Ends Comparison. We first examine the two endpoints $t \rightarrow 0$ and $t \rightarrow 1$. At $t \rightarrow 0$, $\sigma_i(0) = \lambda_i$, recovering the spectrum of Σ_1 ; at $t = 1$, $\sigma_i(1) = 1$ for all i , yielding an isotropic covariance $\Sigma_v(1) = I_d$. To compare these two spectra, we treat $\sigma_i(t) = f_t(\lambda_i)$ as a univariate map with t as a fixed parameter and λ_i as the input variable. A direct computation shows that f_t is strictly increasing, strictly concave, and satisfies $f_t(0^+) = 0$. By Lemma 1 (normalized-spectrum majorization), these three properties imply that the normalized spectrum of $\Sigma_v(t)$ is *majorized* by that of Σ_1 —that is, the concave map f_t compresses large singular values and lifts small ones, producing a strictly flatter output sequence. Consequently, every Schur-concave flatness functional (including r_N) satisfies $r_N(\Sigma_v(t)) \geq r_N(\Sigma_1)$ for any $t \in (0, 1)$, with strict inequality whenever Σ_1 has a non-degenerate spectrum. Detailed arguments for these two endpoints are given in Appendices B.2 and B.3.

Having established flattening relative to the endpoint $t = 0$, we next strengthen the result to *full-range monotonicity*: the normalized spectrum $\{\sigma_i(t)\}_{i=1}^d$ becomes strictly flatter as t increases on the entire interval $[0, 1]$, not merely at the endpoints. This is established in the following subsection by a second application of Lemma 1 to the transition map between arbitrary time pairs $t_1 < t_2$.

Full-Range Monotonicity. We now strengthen Theorem 1 from endpoint comparison to full-range monotonicity. Fix $0 \leq t_1 < t_2 \leq 1$ and define the transition map $h_{t_1 \rightarrow t_2} : (0, \infty) \rightarrow (0, \infty)$ that sends each singular value at time t_1 to its counterpart at time t_2 :

$$h_{t_1 \rightarrow t_2}(s) \triangleq f_{t_2}(f_{t_1}^{-1}(s)), \quad s \in \{\sigma_i(t_1)\}_{i=1}^d, \quad (4)$$

so that $\sigma_i(t_2) = h_{t_1 \rightarrow t_2}(\sigma_i(t_1))$ for every i .

A direct calculation (Appendix B.4) shows that $h_{t_1 \rightarrow t_2}$ is strictly increasing, strictly concave, and satisfies $h_{t_1 \rightarrow t_2}(0^+) = 0$. Applying Lemma 1 to $h_{t_1 \rightarrow t_2}$ with input sequence $\{\sigma_i(t_1)\}_{i=1}^d$ yields

$$\widehat{\sigma}(t_1) \succ \widehat{\sigma}(t_2), \quad (5)$$

where $\widehat{\sigma}(t) = \sigma(t)/\sum_j \sigma_j(t)$ denotes the normalized spectrum. Since every Schur-concave flatness functional respects the majorization order, and r_N is Schur-concave on the simplex of normalized spectra, we conclude

$$r_N(\Sigma_v(t_1)) \leq r_N(\Sigma_v(t_2)), \quad \forall 0 \leq t_1 < t_2 \leq 1, \quad (6)$$

with strict inequality whenever Σ_1 has a non-degenerate spectrum. This completes the proof of Theorem 1.

In the Fig. 2, the r_N line using SVD method to decompose $G^\natural$, and the sim score is calculated by similarity with the actual fine-tuning delta weights subtracted by the final fine-tuning weights and pre-training weights.

Sec. 3.2 characterizes the spectrum of the input-space conditional covariance $\Sigma_v(t)$, which encodes the posterior uncertainty over x_0 given x_t . The empirical gradient that drives spectral initialization, however, is the weight gradient $\hat{G}_t = \frac{1}{N} \sum_i e_t^{(i)} (h_t^{(i)})^\top$, where $e_t^{(i)}$ is the per-sample prediction error and $h_t^{(i)}$ is the backbone feature. The effective rank of $\hat{G}_t$ is therefore governed by the diversity of $\{h_t^{(i)}\}_{i=1}^N$ across samples, rather than by $\Sigma_v(t)$ alone. The two quantities nonetheless align at the data endpoint: as $t \rightarrow 0$, inputs $x_t \rightarrow x_0$ concentrate near the data manifold, so backbone features collapse to a low-dimensional response region and $\text{rank}_{\text{eff}}(\hat{G}_t)$ drops in tandem with the sharpening of $\Sigma_v(t)$. Sec. 3.2 thus provides the input-level justification for the data-side endpoint of our Principal Timesteps; the symmetric low-rank behavior at the noise-side endpoint $t \rightarrow 1$ arises from a complementary mechanism (feature collapse under near-isotropic inputs) that we examine empirically in Section 1.

3.3 Golden Channel

Though $r_N^{G^\natural}$ has decreased to a level that can be reconstructed well by SVD under the LoRA rank budget, two concerns remain. First, the irrelevant information carried by x_t may still contaminate $G^\natural$, so the leading singular directions of $G^\natural$ are not guaranteed to align with the true optimization direction of the fine-tuning trajectory. Second, since the gradients we use for reinitialization are estimated from a finite sample of x_t , SVD—which fits the dominant energy of *this particular sample*—is sensitive to sample-specific noise rather than to the underlying shared signal across samples.

From SVD to PMD. To address both issues, we replace SVD with Penalized Matrix Decomposition (PMD) [41] when extracting the low-rank subspace of $G^\natural$. PMD reformulates the leading singular component as a constrained bilinear maximization with additional sparsity penalties on the factors:

$$\max_{u,v} u^\top G v \quad \text{s.t.} \quad \|u\|_2 \leq 1, \|v\|_2 \leq 1, \|u\|_1 \leq c_1, \|v\|_1 \leq c_2, \quad (7)$$

and subsequent components are obtained by deflation. When c_1, c_2 are large enough, Eq. 7 reduces to the variational form of SVD; when they are tightened, the recovered factors are encouraged to concentrate on a small set of coordinates rather than spreading energy uniformly across all dimensions. This sparsity prior is well aligned with our setting: the true fine-tuning direction is expected to live on a low-rank *and* structurally compact subspace of the parameter space, while the noise introduced by irrelevant x_t tends to be diffuse. By penalizing diffuse factors, PMD is biased away from sample-specific fluctuations and toward directions that are consistently expressed across gradient samples. We therefore expect PMD to yield a $G^\natural$ that is (i) better aligned with the large-sample limit and (ii) more robust to perturbations on individual samples.

Empirical verification. We validate the golden channel with the experiment summarized in Fig. 2. We treat the gradient estimated from a large number of x_t samples (50000 iters) as a proxy for the true optimization direction, and measure how well the rank- r subspace recovered from a small-sample gradient aligns with it, using the arc sine of the principal angle (smaller is better). We compare two decompositions, SVD and PMD, in two regimes: clean small-sample gradients, and under two kinds of noise (Gaussian noise sampled from a Gaussian distribution and random label noise sampled from another dataset).

Two observations support the use of PMD. (i) *Better alignment.* Across all sample budgets, the PMD subspace is markedly closer to the large-sample direction than the SVD subspace; for instance, with only 10 iters PMD already attains a closer angle compared with the 10^4 sampled SVD result. (ii) *Better noise robustness.* Under additive Gaussian and sample from another dataset perturbation, SVD degrades substantially, whereas PMD remains nearly unchanged, indicating that the PMD subspace is governed by the cross-sample shared signal rather than by sample-specific fluctuations.

Together, these results confirm that PMD recovers a $G^\natural$ that is both more faithful to the true fine-tuning direction and more stable under noise—exactly the properties we need for a reliable reinitialization of LoRA.

3.4 LoRA Initialization via Spectral Decomposition

The training pipeline of PRISM-LoRA consists of three stages: *gradient estimation*, *parameter initialization*, and *fine-tuning*. We will introduce the process of each stage in the following.

Stage 1: Gradient Estimation. In the first stage, we collect M gradient samples by performing forward-backward passes under the principal timestep distribution:

$$G^\natural = \frac{1}{M} \sum_{i=1}^M \nabla_{\theta} \mathcal{L}_{\text{FM}}(\theta; x_0^{(i)}, x_1^{(i)}, t_i), \quad t_i \sim p^\natural(t), \quad x_0^{(i)} \sim \mathcal{N}(0, I), \quad x_1^{(i)} \sim p_{\text{data}}, \quad (8)$$

where θ denotes the (frozen) base-model parameters of the target weight matrix, M is the number of gradient samples, $\mathcal{L}_{\text{FM}}$ is the flow-matching loss defined in Eq. (10), $p^\natural(t)$ is the principal timestep distribution supported on a low- t subinterval of $[0, 1]$, $x_0^{(i)}$ and $x_1^{(i)}$ are the noise and data endpoints of the i -th sample, and $G^\natural$ is the resulting low-rank gradient estimate on which the subsequent decomposition operates.

Stage 2: Parameter Initialization. In the second stage, we apply the golden-channel sparse SVD to $G^\natural$ and initialize the LoRA factors from its leading spectral components:

$$A_0 = \sqrt{\gamma} [U_{G^\natural}]_{[:, 1:r]} [S_{G^\natural}^{1/2}]_{[1:r]}, \quad B_0 = \sqrt{\gamma} [S_{G^\natural}^{1/2}]_{[1:r]} [V_{G^\natural}]_{[:, 1:r]}^\top, \quad (9)$$

where $U_{G^\natural}$, $S_{G^\natural}$, and $V_{G^\natural}$ are the left singular vectors, singular values, and right singular vectors produced by the golden-channel sparse decomposition of $G^\natural$; r is the target LoRA rank; $[:, 1:r]$ and $[1:r, :]$ denote truncation to the top- r components; γ is the LoRA scaling factor; and $A_0 \in \mathbb{R}^{r \times d_{\text{out}}}$ and $B_0 \in \mathbb{R}^{d_{\text{in}} \times r}$ are the initialized LoRA down- and up-projection matrices, respectively.

Stage 3: Fine-tuning. In the final stage, we fine-tune the LoRA factors (A, B) initialized from Eq. (9) using the standard flow-matching objective, without any further modification to the timestep distribution:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(0,1), x_0 \sim \mathcal{N}(0,I), x_1 \sim p_{\text{data}}} \left[\|v_\theta(x_t, t) - (x_1 - x_0)\|_2^2 \right], \quad x_t = (1-t)x_0 + tx_1, \quad (10)$$

where $v_\theta(\cdot, \cdot)$ is the flow-matching velocity predictor parameterized by the base weights and the trainable LoRA factors, $t \sim \mathcal{U}(0, 1)$ is the training timestep sampled uniformly over the full interval, x_t is the linear interpolant between noise x_0 and data x_1 , and $x_1 - x_0$ is the target velocity field. During this stage, the base parameters are kept frozen, and only (A, B) are updated.

Table 1: **Quantitative comparison with existing methods** on different text-to-image generation tasks. The **bold** and underlined figures represent the optimal and sub-optimal results, respectively.

Task	Methods / Setting	Controllability		Alignment			Image Quality		
		F1 $\uparrow$	MSE $\downarrow$	CLIP Text $\uparrow$	CLIP Image $\uparrow$	DINO $\uparrow$	FID $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$
Dreambooth	SD3-medium	-	-	<u>0.224</u>	0.793	<u>0.667</u>	170.0	-	-
	+ Pissa	-	-	0.104	0.475	0.024	418.0	-	-
	+ LoRA-GA	-	-	0.168	0.699	0.533	260.0	-	-
	+ LoRA-One	-	-	<u>0.224</u>	<u>0.795</u>	<u>0.667</u>	<u>169.1</u>	-	-
	+ PRISM-LoRA	-	-	0.230	0.805	0.682	160.3	-	-
Canny	OminiControl	0.4763	-	0.2203	0.7085	0.5011	97.56	0.3702	8.57
	+ Pissa	0.0588	-	0.1466	0.4964	-0.0166	456.62	0.2297	5.67
	+ LoRA-GA	0.4305	-	0.2128	0.6802	0.4552	108.25	0.2397	7.34
	+ LoRA-One	<u>0.4893</u>	-	0.2232	<u>0.7214</u>	<u>0.5339</u>	<u>95.93</u>	0.3618	<u>8.73</u>
	+ PRISM-LoRA	0.4978	-	<u>0.2227</u>	0.7242	0.5471	95.38	<u>0.3311</u>	8.82
Depth	OminiControl	-	904.9	0.2082	<u>0.6601</u>	0.4470	101.88	0.3376	7.91
	+ Pissa	-	9416	0.1650	0.5233	-0.0034	500.50	0.3140	8.46
	+ LoRA-GA	-	12160	0.0766	0.4959	0.1319	283.29	0.2333	9.67
	+ LoRA-One	-	1117	0.2022	0.6491	0.3999	119.16	0.3287	8.01
	+ PRISM-LoRA	-	851.5	<u>0.2074</u>	0.7587	0.3939	127.66	<u>0.3317</u>	<u>9.45</u>
Deblur	OminiControl	-	89.39	0.2518	0.8083	<u>0.6660</u>	<u>91.47</u>	0.4835	15.99
	+ Pissa	-	91.43	0.2497	0.7932	0.6465	102.0	0.4731	15.64
	+ LoRA-GA	-	85.24	0.2517	0.8137	0.6575	91.70	0.4840	15.90
	+ LoRA-One	-	<u>69.07</u>	<u>0.2528</u>	<u>0.8203</u>	0.6589	92.72	<u>0.4913</u>	<u>16.53</u>
	+ PRISM-LoRA	-	55.87	0.2535	0.8328	0.7232	77.85	0.5224	17.32

4 Experiment

Experimental Setting. We evaluate our method against sota methods across different image generation tasks: subject-driven generation, spatially-aligned generation, image deblurring, etc. To

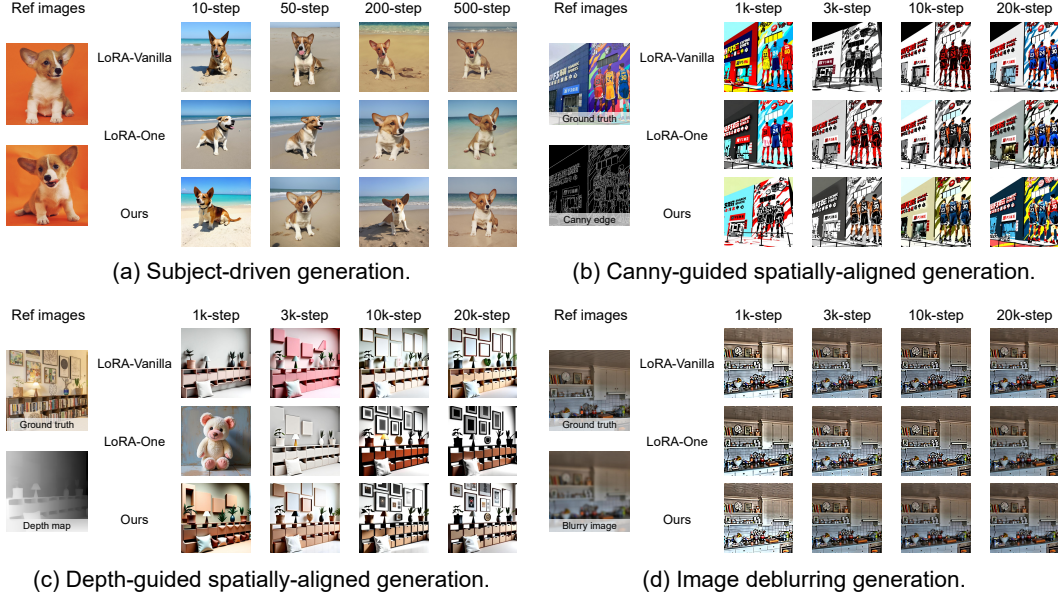


Figure 3: **Visual comparison with sota methods** across various text-to-image generation tasks based on LoRA fine-tuning. Our method more rapidly aligns with the target distribution, and optimizes the final generation performance. *Caption: A sks dog on the beach. A high resolution picture of building. A high resolution picture of in-door decoration. A high resolution picture of kitchen.*

prove PRISM-LoRA can be effective across models, we deploy PRISM-LoRA on two mainstream models, FLUX-1 [18] and Stable Diffusion-3 [7]. Regarding dataset setting, we use the last 100,000 images in the MultiGen-20M [50] dataset for training, and last 2,500 images in COCO for evaluation in Canny, Depth, and Deblur tasks. For Dreambooth, we use the original 30 categories of images in training and generate 4 images with 25 prompts, with a total of 3,000 images for evaluation. See more details about hyperparameter and other experiment settings in Appendix. C.

We compare our method with other spectral-init methods, Pissa [25], LoRA-GA [39], and LoRA-One [47]. In the spatial alignment task, we use OminiControl [35] as the baseline method, which uses the vanilla LoRA initialization method, and deploy the above spectral-init methods.

Evaluation Metrics. We evaluate generated images along three axes. Controllability measures fidelity to the input condition: F1 score between Canny edges extracted from the generated and input images for the Canny task, and MSE between predicted and input depth maps for the Depth task. Alignment assesses semantic consistency via CLIP Text (text-image similarity), CLIP Image (image-image similarity), and DINO [2] feature similarity against the reference. Image quality is reported in terms of FID [11], SSIM, and PSNR. Further details are provided in Appendix. C.

4.1 Main Results

Quantitative comparison. Table 1 summarizes the performance across the four evaluated tasks. Our proposed PRISM-LoRA demonstrates superior controllability across all spatially-aligned and restoration tasks, achieving peak performance with an F1 score of 0.4978 on Canny, and minimizing MSE to 851.5 on Depth and 55.87 on Deblur. Notably, on the Deblur task, PRISM-LoRA consistently outperforms the OmniControl baseline across nearly all metrics, significantly improving FID from 91.47 to 77.85 and PSNR from 15.99 to 17.32. For subject-driven generation (DreamBooth), our method establishes robust leadership among LoRA variants in both semantic alignment (CLIP Text: 0.230, CLIP Image: 0.805, DINO: 0.682) and image quality (FID: 160.3), and achieve an average of 557 steps to reach best score, compared with LoRA-Base with 650, LoRA-GA 957 and LoRA-One 757 steps. Furthermore, while improvements in alignment and fidelity on Canny and Depth are relatively modest, PRISM-LoRA remains highly competitive, consistently matching or surpassing existing spectral-init variants. It is also crucial to note that PiSSA experiences severe performance degradation on most tasks (e.g., yielding an FID of 456.62 on Canny and 500.50 on Depth). This

stark contrast underscores the critical importance of gradient initialization quality and confirms that PRISM-LoRA provides a fundamentally more robust optimization starting point.

Qualitative comparison. Fig. 3 provides visual comparisons across the four downstream tasks at various training milestones. Across all settings, PRISM-LoRA exhibits a substantially accelerated convergence rate compared to both vanilla LoRA and LoRA-One. Specifically, coherent semantic structures emerge at remarkably early iterations (e.g., 10–500 steps), whereas the baselines continue to yield blurry, misaligned, or content-collapsed outputs at the equivalent stages. In the DreamBooth task, PRISM-LoRA faithfully preserves the unique identity of the reference subject (the specific dog) throughout the entire training trajectory. In contrast, vanilla LoRA suffers from concept drift toward generic dogs, and LoRA-One struggles to recover the subject identity until much later steps. For spatially-aligned generation (Canny and Depth), our model adheres more rigorously to the input conditioning signals, delivering sharper edge boundaries and highly accurate spatial layouts. Finally, in the image deblurring task, PRISM-LoRA not only restores high-frequency details earlier in the training process but also achieves perceptually cleaner reconstructions at convergence.

Table 2: **Ablation experiments** on Dreambooth and Canny-guided controllable generation.

Tasks	Methods	F1↑	FID↓	CLIP-I↑	DINO↑
Dreambooth	LoRA-One	-	169.1	0.795	0.667
	w. principal timestep	-	161.7	0.798	0.677
	w. golden channel	-	158.9	0.800	0.674
	PRISM-LoRA	-	160.3	0.805	0.682
Canny	LoRA-One	0.4893	95.93	0.7214	0.5339
	w. principal timestep	0.4834	95.82	0.7135	0.5351
	w. golden channel	0.4912	97.14	0.7158	0.5387
	PRISM-LoRA	0.4978	95.38	0.7242	0.5471

4.2 Ablation Study

To isolate the individual contributions of our proposed modules—Principal Timestep selection and Golden Channel filtering—we evaluate their independent and combined effects on the Dreambooth (subject-driven generation) and Canny (spatially-aligned generation) tasks. As observed in Table. 2, integrating either the principal timestep or the golden channel independently into the strong LoRA-One baseline yields overall performance improvements, particularly in terms of identity preservation and image quality. On the Dreambooth task, applying the Golden Channel alone improves the DINO score from 0.667 to 0.674 and substantially reduces FID, suggesting that filtering out task-irrelevant channels is crucial for maintaining optimization stability even under stochastic timestep sampling.

While individual components exhibit distinct merits, their combination in the full PRISM-LoRA framework achieves the optimal performance balance. Interestingly, on the Canny task, while applying independent modules causes slight fluctuations in specific metrics, the complete PRISM-LoRA clearly dominates, achieving the highest F1 score (0.4978) and DINO score (0.5471). This demonstrates a strong synergistic effect: suppressing the effective gradient rank (via Principal Timestep) creates a cleaner spectral foundation, which enables the sparse decomposition (via Golden Channel) to isolate the optimal fine-tuning directions more accurately. Together, they successfully mitigate the high-rank gradient dilemma inherent in flow-matching diffusion models.

5 Conclusion

In this paper, we identify and resolve a fundamental mismatch between the low-rank assumption of standard spectral initialization and the intrinsically high-rank gradient geometry found in flow-matching diffusion models. We demonstrate that stochastic timestep sampling induces directionally heterogeneous gradients, which renders conventional low-rank approximations lossy and misaligned. To overcome this barrier, we propose PRISM-LoRA, a principled spectral initialization framework specifically tailored for diffusion fine-tuning. By synergizing Principal Timestep selection to inherently suppress the effective gradient rank, and Golden Channel sparse decomposition to filter out task-irrelevant noise, PRISM-LoRA ensures that the initialized parameters robustly align with the long-horizon optimization trajectory.

Limitation and future works While our approach achieves both faster convergence and strong performance on this task, a more comprehensive validation across diverse subject-driven scenarios remains an interesting direction for future work. On the theoretical side, our analysis primarily focuses on the observation that oversampling intermediate timesteps can be harmful to gradient quality. However, these timesteps also exhibit the largest variance, suggesting that they carry informative gradient signals that should be properly trained and accelerated rather than down-weighted. Extending our theory to characterize the spectral behavior across the full range of diffusion timesteps is a promising avenue for future investigation.

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Appendix

A Preliminaries

A.1 Flow Matching Models

Prior diffusion models based on denoising diffusion probabilistic models (DDPMs) [12] learn generative processes by gradually adding Gaussian noise to the data and training a model to reverse this stochastic diffusion process. In contrast, Flow Matching (FM) [21] formulates generative modeling as learning a continuous-time velocity field that deterministically transports samples from a simple prior distribution to the data distribution. Let $x_0 \sim p_0(x)$ and $x_1 \sim q(x)$ denote noise and data samples. The interpolation path is as follows:

$$x_t = (1 - t)x_0 + tx_1. \quad (11)$$

As above, a vector field $v_\theta(x, t)$ —parameterized by a neural network, is constructed to approximate a target vector field $v_t(x)$, which transports a simple prior distribution $p_0(x)$ to a complex data distribution $p_1(x) \approx q(x)$ via an ordinary differential equation (ODE): $\frac{d}{dt}\phi_t(x) = v_t(\phi_t(x))$ with $\phi_0(x) = x$. The ideal marginal FM objective is defined as:

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0,1], x \sim p_t(x)} \left[\|v_\theta(x, t) - v_t(x)\|_2^2 \right]. \quad (12)$$

However, the marginal probability path $p_t(x)$ and the vector field $v_t(x)$ are generally unknown. To bypass this intractability, CFM [19] constructs the target vector field by conditioning on individual data samples $x_1 \sim q(x_1)$. Given a conditional probability path $p_t(x|x_1)$ and its generating vector field $v_t(x|x_1)$, the tractable CFM objective is:

$$\begin{aligned} \mathcal{L}^*(\theta) &= \mathbb{E}_{t, x_1 \sim q(x_1), x \sim p_t(x|x_1)} \left[\|v_\theta(x, t) - v_t(x|x_1)\|_2^2 \right] \\ &= \mathbb{E}_{t, x_1 \sim q(x_1), x \sim p_t(x|x_1)} \left[\|v_\theta((1-t)x_0 + tx_1, t) - (x_1 - x_0)\|_2^2 \right] \end{aligned} \quad (13)$$

Minimizing $\mathcal{L}_\theta^*$ is mathematically equivalent to minimizing the intractable $\mathcal{L}_\theta$ up to a constant. The inference process solves the ODE backward from $t = 0$ to $t = 1$ through n iterative updates:

$$x_{t_{i+1}} = x_{t_i} - (t_i - t_{i+1})v_\theta(x_{t_i}, t_i), \quad x_{t_0} \sim \mathcal{N}(0, I), \quad (14)$$

with final output x_{t_n} as the generated sample.

A.2 Gradient-guided Parameter Efficient Fine-tuning

Low-Rank Adaptation [13] is one of the parameter efficient fine-tuning methods that adapts large pre-trained models by injecting trainable low-rank matrices into existing weight layers. Given a pre-trained weight matrix $W \in \mathbb{R}^{d \times k}$, LoRA models its update as a low-rank decomposition:

$$W' = W + \Delta W, \quad \Delta W = \eta B A, \quad (15)$$

where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$, $r \ll \min(d, k)$ denotes the rank, and η is a scaling factor. This significantly reduces the number of trainable parameters and memory footprint.

Theoretically, the update direction of the LoRA module can be aligned with the full fine-tuning at the early training stage via the first gradient descent step, which can be mathematically expressed as:

$$\eta(\Delta B A_{\text{init}} + B_{\text{init}} \Delta A) = \eta \lambda [\nabla_B \mathcal{L}(B_{\text{init}}) A_{\text{init}} + B_{\text{init}} \nabla_A \mathcal{L}(A_{\text{init}})], \quad (16)$$

To measure its approximation quality of scaled the update of the weights in full fine-tuning $\zeta \Delta W = \zeta \lambda \nabla_W \mathcal{L}(W_0)$, the Frobenius norm of the difference between these two updates [39, 47] is commonly used as a criterion:

$$\begin{aligned} &\|\eta(\Delta B A_{\text{init}} + B_{\text{init}} \Delta A) - \zeta \lambda \nabla_W \mathcal{L}(W_0)\|_F \\ &= \lambda \|\eta \nabla_B \mathcal{L}(B_{\text{init}}) A_{\text{init}} + \eta B_{\text{init}} \nabla_A \mathcal{L}(A_{\text{init}}) - \zeta \nabla_W \mathcal{L}(W_0)\|_F \\ &= \lambda \|\eta^2 \nabla_{W'} \mathcal{L}(W_0) \cdot A_{\text{init}}^T A_{\text{init}} + \eta^2 B_{\text{init}} B_{\text{init}}^T \cdot \nabla_W \mathcal{L}(W_0) - \zeta \nabla_W \mathcal{L}(W_0)\|_F. \end{aligned} \quad (17)$$

The svd decomposition can be formulated as $G^\natural = U_{G^\natural} S_{G^\natural} V_{G^\natural}$. Matrix U and V are orthogonal matrix, and S is a strictly increasing sequence of singular value formulated as $S_{G^\natural} = \{\sigma_1, \sigma_2, \dots, \sigma_d\}$.

B Detailed proof of principal timesteps

B.1 Problem Formulation

Consider the linear interpolation $x_t = (1-t)x_0 + tx_1$, where $x_0 \sim \mathcal{N}(0, I_d)$ is independent of x_1 . Throughout this section, we adopt the *Gaussian surrogate assumption* $x_1 \sim \mathcal{N}(0, \Sigma_1)$, which yields closed-form conditional covariances; for non-Gaussian p_{data} the expressions below serve as the best linear-Gaussian approximation of $\text{Var}[x_0 | x_t]$. Let the eigenvalues of Σ_1 be

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d > 0,$$

and assume $\lambda_1 > \lambda_d$ (non-degenerate spectrum). The (rescaled) gradient covariance matrix is defined as

$$\Sigma_v(t) = \frac{1}{t^2} \text{Var}[x_0 | x_t]. \quad (18)$$

To obtain Eq. 18, we need to first make a simple conversion for x_1 . According to the flow-matching identity, we can express x_1 as $x_1 = \frac{x_t - (1-t)x_0}{t}$. As a result, the model prediction $v(x_t, t)$ can be express as

$$v(x_t, t) = \mathbb{E}[x_1 - x_0 | x_t] = \mathbb{E}\left[\frac{x_t - x_0}{t} | x_t\right] = \frac{x_t - \mathbb{E}[x_0 | x_t]}{t}. \quad (19)$$

From this expression, we observe that the velocity field is determined by posterior expectation. Finally, with the irreducible noise $x_1 - x_0 - v(x_t, t) = \frac{\mathbb{E}[x_0 | x_t] - x_0}{t}$, the covariance of the irreducible noise is $\text{Cov}\left(\frac{\mathbb{E}[x_0 | x_t] - x_0}{t} | x_t\right) = \frac{1}{t^2} \text{Var}[x_0 | x_t]$.

Derivation of the spectrum. Because $x_0 \perp x_1$, the marginal covariance of x_t is $\text{Cov}(x_t) = (1-t)^2 I_d + t^2 \Sigma_1$, and the cross-covariance is $\text{Cov}(x_0, x_t) = (1-t)I_d$. Under the joint-Gaussian assumption, the conditional covariance is

$$\text{Var}[x_0 | x_t] = I_d - (1-t)^2 [(1-t)^2 I_d + t^2 \Sigma_1]^{-1}. \quad (20)$$

Diagonalizing in the eigenbasis of Σ_1 , the i -th eigenvalue of $\text{Var}[x_0 | x_t]$ equals $\frac{t^2 \lambda_i}{(1-t)^2 + t^2 \lambda_i}$, so the eigenvalues $\sigma_i(t)$ of $\Sigma_v(t)$ satisfy

$$\sigma_i(t) = \frac{\lambda_i}{(1-t)^2 + t^2 \lambda_i}. \quad (21)$$

B.2 Spectral Comparison at t in $\{0, 0.5, 1\}$

Evaluating (21) at characteristic timesteps:

- **At $t \rightarrow 0$:** $\sigma_i(0) = \lambda_i$. The spectrum coincides with the data covariance Σ_1 .
- **At $t = 0.5$:** $\sigma_i(0.5) = \frac{4\lambda_i}{1 + \lambda_i}$.
- **At $t = 1$:** $\sigma_i(1) = 1$ for all i , so $\Sigma_v(1) = I_d$. Intuitively, when $x_t = x_1$ carries no information about x_0 , the conditional variance reduces to the prior I_d and the gradient direction is dominated entirely by noise.

B.3 Majorization Lemma

We first establish a general majorization result from which the three-point comparison (and the full-range monotonicity) will follow.

Lemma 1 (Normalized-spectrum majorization). *Let $\lambda = (\lambda_1, \dots, \lambda_d)$ with $\lambda_1 \geq \dots \geq \lambda_d > 0$ and $\lambda_1 > \lambda_d$. Let $f : (0, \infty) \rightarrow (0, \infty)$ be strictly increasing, strictly concave, and satisfy $f(0^+) = 0$. Define the normalized spectra*

$$\hat{a}_i = \frac{\lambda_i}{\sum_j \lambda_j}, \quad \hat{b}_i = \frac{f(\lambda_i)}{\sum_j f(\lambda_j)}.$$

Then $\hat{a} \succ \hat{b}$ (strictly), i.e.

$$\sum_{i=1}^k \hat{a}_i \geq \sum_{i=1}^k \hat{b}_i \quad \forall k \in \{1, \dots, d\},$$

with strict inequality for at least one $k < d$.

Proof. Both $\hat{a}$ and $\hat{b}$ sum to 1, and since f is strictly increasing, both are non-increasing in i (same ordering). Consider the ratio

$$\frac{\hat{a}_i}{\hat{b}_i} = \frac{\lambda_i}{f(\lambda_i)} \cdot \frac{\sum_j f(\lambda_j)}{\sum_j \lambda_j}. \quad (22)$$

Strict concavity of f together with $f(0^+) = 0$ implies that $\lambda \mapsto \lambda/f(\lambda)$ is strictly increasing on $(0, \infty)$.³ Since λ_i is non-increasing in i , the ratio $\hat{a}_i/\hat{b}_i$ is non-increasing in i ; because $\lambda_1 > \lambda_d$, it is not constant.

Cut-property argument. Since $\hat{a}_i/\hat{b}_i$ is non-increasing and non-constant, and $\sum_i \hat{a}_i = \sum_i \hat{b}_i = 1$, there exists a unique index $i^* \in \{1, \dots, d-1\}$ such that

$$\hat{a}_i \geq \hat{b}_i \text{ for } i \leq i^*, \quad \hat{a}_i \leq \hat{b}_i \text{ for } i > i^*.$$

For $k \leq i^*$, each summand in $\sum_{i=1}^k (\hat{a}_i - \hat{b}_i)$ is non-negative, so the partial sum is ≥ 0 . For $k > i^*$, using $\sum_{i=1}^d (\hat{a}_i - \hat{b}_i) = 0$,

$$\sum_{i=1}^k (\hat{a}_i - \hat{b}_i) = - \sum_{i=k+1}^d (\hat{a}_i - \hat{b}_i) \geq 0, \quad (23)$$

since each term in the tail sum is ≤ 0 . Hence $\sum_{i=1}^k \hat{a}_i \geq \sum_{i=1}^k \hat{b}_i$ for all k , with strict inequality at $k = i^*$. $\square$

B.4 Full-Range Monotonicity of Spectral Flatness

We now promote the three-point comparison to full-range monotonicity in t .

Lemma 2 (Monotone majorization along t). *For any $0 \leq s < t \leq 1$, let $\hat{\sigma}(s)$ and $\hat{\sigma}(t)$ denote the normalized spectra of $\Sigma_v(s)$ and $\Sigma_v(t)$ respectively. Then $\hat{\sigma}(s) \succ \hat{\sigma}(t)$.*

Proof. Define, for fixed $t \in [0, 1]$, the map $g_t : (0, \infty) \rightarrow (0, \infty)$ by $g_t(\lambda) = \frac{\lambda}{(1-t)^2 + t^2 \lambda}$, so that $\sigma_i(t) = g_t(\lambda_i)$. A direct computation gives

$$g'_t(\lambda) = \frac{(1-t)^2}{[(1-t)^2 + t^2 \lambda]^2} > 0, \quad g''_t(\lambda) = -\frac{2t^2(1-t)^2}{[(1-t)^2 + t^2 \lambda]^3} \leq 0,$$

so g_t is strictly increasing and (for $t \in (0, 1)$) strictly concave; moreover $g_t(0^+) = 0$.

For $s = 0$ and any $t \in (0, 1]$, the claim $\hat{\sigma}(0) \succ \hat{\sigma}(t)$ follows immediately from Lemma 1 applied with $f = g_t$.

For general $0 < s < t \leq 1$, write $\sigma_i(t) = (g_t \circ g_s^{-1})(\sigma_i(s))$. Let $h = g_t \circ g_s^{-1}$. Both g_t and g_s^{-1} are strictly increasing, so h is strictly increasing; and the composition of a concave increasing function with an increasing concave inverse (i.e. with a convex g_s^{-1} — note that the inverse of a concave increasing function is convex increasing) requires care. A direct calculation yields

$$\begin{aligned} h(\mu) &= g_t\left(\frac{(1-s)^2 \mu}{1-s^2 \mu}\right) \\ &= \frac{(1-s)^2 \mu}{(1-t)^2(1-s^2 \mu) + t^2(1-s)^2 \mu} \\ &= \frac{(1-s)^2 \mu}{(1-t)^2 + [t^2(1-s)^2 - s^2(1-t)^2] \mu}. \end{aligned} \quad (24)$$

³Indeed, $\frac{d}{d\lambda}(\lambda/f(\lambda)) = (f(\lambda) - \lambda f'(\lambda))/f(\lambda)^2$; strict concavity and $f(0^+) = 0$ give $f(\lambda) > \lambda f'(\lambda)$ for all $\lambda > 0$.

Setting $\alpha = (1 - s)^2 > 0$ and $\beta = t^2(1 - s)^2 - s^2(1 - t)^2$, we have $h(\mu) = \frac{\alpha\mu}{(1-t)^2 + \beta\mu}$. For $s < t \leq 1$ one checks that $\beta > 0$ ⁴ hence h is of the same Möbius form $\frac{\alpha\mu}{c + \beta\mu}$ with $c, \alpha, \beta > 0$, so h is strictly increasing, strictly concave, and $h(0^+) = 0$.

Applying Lemma 1 to the spectrum $\sigma(s)$ with transformation $f = h$ gives $\hat{\sigma}(s) \succ \hat{\sigma}(t)$. $\square$

B.5 r_N Flatness

Define the r_N metric $r_N(\hat{x}) = \min r : \sum_{i=1}^r \hat{x}_{(i)} \geq N/100$,

where $\hat{x}_{(1)} \geq \hat{x}_{(2)} \geq \dots$ is the sorted spectrum. A standard consequence of majorization is that if $\hat{a} \succ \hat{b}$ then $r_N(\hat{a}) \leq r_N(\hat{b})$, since larger partial sums reach the $N/100$ threshold no later.

Combining this with Lemma 2 yields:

Theorem 2 (Monotone spectral flattening). *Under the assumptions of Section B (Gaussian surrogate, $\lambda_1 > \lambda_d$), the r_N metric is non-decreasing in t on $[0, 1]$: for all $0 \leq s < t \leq 1$, $r_N(\Sigma_v(s)) \leq r_N(\Sigma_v(t))$. In particular,*

$$r_N(\Sigma_v(0)) \leq r_N(\Sigma_v(0.5)) < r_N(\Sigma_v(1)) = \lceil N/100 \rceil, \quad (25)$$

where the strict inequality $r_N(\Sigma_v(0.5)) < r_N(\Sigma_v(1))$ holds because $\hat{\sigma}(1) = (1/d, \dots, 1/d)$ is the uniform distribution, which is strictly majorized by any non-uniform $\hat{\sigma}(0.5)$ whenever $\lambda_1 > \lambda_d$.

This proves that the gradient covariance becomes progressively flatter as t moves from the data distribution toward the noise: small t (the “head” of the sampling trajectory) admits a sharper low-rank structure in its gradient spectrum, whereas large t (the “tail”) approaches an isotropic spectrum and is inherently high-rank. This monotonicity motivates treating small- t timesteps as the *golden timesteps* for low-rank gradient-based initialization.

C Experimental Settings and Additional Results

Evaluation Metrics We evaluate our method from three complementary perspectives: controllability, alignment, and image quality. For **controllability**, we measure how faithfully the generated images respect the input control signal. On the Canny task, we extract Canny edges from the generated images and compute the F1 score against the input edge map, where higher values indicate better edge preservation. On the Depth task, we estimate the depth map of the generated images using a pretrained depth predictor and report the Mean Squared Error (MSE) against the input depth condition, where lower values indicate more accurate geometric alignment.

For **alignment**, we assess semantic consistency from both textual and visual perspectives. *CLIP Text* computes the cosine similarity between CLIP embeddings of the generated image and the text prompt, reflecting text-image semantic alignment. *CLIP Image* measures the cosine similarity between CLIP embeddings of the generated and reference images, capturing high-level visual consistency. *DINO* reports the cosine similarity in the DINO [2] feature space, which is more sensitive to fine-grained structural and identity-level correspondence than CLIP.

For **image quality**, we adopt three widely-used metrics, and calculated by pyiqa [3]. *FID* measures the distributional distance between generated and real images in the Inception feature space, with lower values indicating more realistic generations. *SSIM* evaluates structural similarity between the generated image and the reference at the pixel level, accounting for luminance, contrast, and structural information. *PSNR* measures pixel-wise reconstruction fidelity in decibels, where higher values denote less distortion. Together, these metrics provide a comprehensive assessment of perceptual realism, structural consistency, and reconstruction accuracy.

C.1 Implementation Details

In this section, we briefly introduce the datasets, training details, and evaluation metrics used in experiments.

⁴ $\beta = [t(1 - s) - s(1 - t)][t(1 - s) + s(1 - t)] = (t - s)[t(1 - s) + s(1 - t)] > 0$ for $0 \leq s < t \leq 1$.

Hyperparameters setting We implement our method on top of two baselines. For the OminiControl [35] baseline (Canny-to-image, depth-to-image, and deblurring), we follow the original setup: FLUX.1 as the base DiT, LoRA rank 4, effective batch size 8 (batch size 1 with gradient accumulation of 8), prodigy optimizer with weight decay 0.01, and 512×512 resolution. Training uses the last 300,000 images of text-to-image-2M, with Canny/depth maps and Gaussian blur generated on the fly. For the DreamBooth [32] baseline we use Stable Diffusion 3-medium with the default DreamBooth setting: 30 subjects with 3–5 reference images each, prompts of the form “*a [sks] [class noun]*”, class-specific prior-preservation loss with $\lambda = 1$ (~ 1000 class samples), and learning rate 5×10^{-6} . Training iterations for each LoRA variant are reported in Table 1.

For our method hyperparameters, we use 1000 to 800 and 100 to 0 as our timestep selection to better fit the low-rank hypothesis. For golden channel, we set the L_1 normalized bound on 1, indicating the most sparse sampling situation. For rank selection after sparse iteration, we select the top-2 rank of singular values to remain instead of being set to 0.

We deploy our models on NVIDIA L40S GPU, with 48 gigabytes of VRAM. For dreambooth, the total training time is 25 minutes per class; for the subject-alignment task, we use 2 L40S and train for almost 20 hours.

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